

Smart Metro Ticket Booking System

PROJECT DEVELOPMENT PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. User Acceptance Testing

2.1 Objective

The objective of User Acceptance Testing (UAT) is to confirm that the Smart Metro Ticket Booking System meets the real-world needs of its intended users, daily commuters, students, senior citizens, and differently-abled passengers, by having representative end users book tickets, calculate fares, receive QR-code tickets, and review their booking history through the ServiceNow Service Catalog before the solution is signed off for release.

2.2 Scope of Testing

- End-to-end ticket booking journey through the Service Catalog item, from station selection to confirmation.
- Accuracy and clarity of the automatically calculated fare shown to the user before payment.
- Usability and readability of the QR-code digital ticket delivered after booking.
- Timeliness and clarity of SMS/Email booking confirmation notifications.
- Ability of users to view their booking history and status through their Service Catalog request record.
- Overall ease of use for first-time users, including elderly and less tech-familiar commuters.

2.3 UAT Approach

A group of 15 volunteer participants, representing daily commuters, students, senior citizens, and occasional travellers, tested the system on the ServiceNow sub-production instance over a two-day cycle. Each participant followed a set of predefined test cases and recorded their observations, along with any usability issues or unexpected behaviour encountered during the booking journey.

User Acceptance Testing Flow



Figure 1: End-to-end User Acceptance Testing process flow

2.4 UAT Test Cases

Test Case	Description	Expected Outcome
Book a ticket	User selects source, destination, date, and passenger count	Booking is created without errors
View fare before payment	User checks calculated fare prior to confirming booking	Correct fare is shown instantly
Receive digital ticket	User completes booking and receives QR-code ticket	QR ticket delivered within seconds
Receive confirmation	User checks SMS/Email after successful booking	Confirmation received promptly
View booking history	User opens their past requests in the Service Catalog	All past bookings listed correctly

2.5 UAT Results Summary

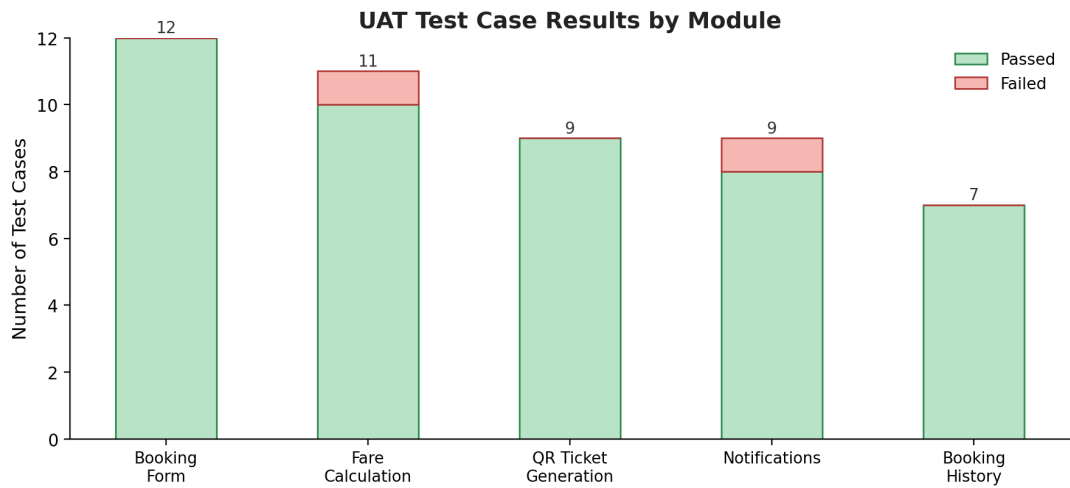


Figure 2: UAT test case results by functional module

2.6 Feedback Summary

Area	Participant Feedback
Booking Form	Rated simple and quick; most users completed booking in under a minute.
Fare Calculation	Users appreciated seeing the fare instantly before confirming payment.
QR Ticket	Well received; senior citizens suggested a larger QR display option.
Notifications	Confirmation messages were clear; one delay reported during peak testing.
Accessibility	First-time and elderly users found the guided catalog form easy to follow.

2.7 Key Observations

- 14 of 15 participants completed the full booking journey successfully on their first attempt.
- The automated fare calculation was consistently rated as fast, accurate, and easy to understand.
- The QR-code digital ticket was well received, with a minor suggestion to enlarge the QR display for elderly users.
- One participant reported a delayed SMS notification during a simulated peak-load test window.

- Overall satisfaction score from participants averaged 4.6 out of 5 for the booking experience.

2.8 Defects and Recommendations

- Minor: Increase the default QR-code display size for improved scannability by elderly users.
- Minor: Investigate the isolated SMS notification delay observed under simulated peak load.
- No critical or high-severity defects were identified that block release.

2.9 Sign-off

Based on the results of this UAT cycle, the Smart Metro Ticket Booking System satisfies the core functional and usability requirements of its target commuters. With the two minor recommendations noted above tracked for a future release, the system is accepted by the UAT participant group as ready to proceed toward deployment.